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PROJEKTE

Dynamic Content Generation System with Retrieval-Augmented Processing [🔗](#)

Tech stack: Python, Scrapy, LangChain, OpenAI APIs, Hugging Face embeddings, FAISS or Pinecone, Streamlit

- Developed a web scraping and preprocessing pipeline to extract and structure AW-Wiki content from Bad Breisig for downstream retrieval.
- Implemented a RAG architecture with embedding-based indexing and semantic search for context-grounded response generation.
- Optimized chunking and retrieval flow to improve answer relevance, response quality, and modular scalability.

Aircraft Engine Remaining Useful Life (RUL) Prediction Using LSTM [🔗](#)

- Built an LSTM-based time-series model to predict engine Remaining Useful Life using the NASA Turbofan dataset.
- Engineered sequence features and trained the model on multi-sensor cycle data to capture degradation patterns.
- Evaluated forecasting performance and summarized insights for predictive maintenance and anomaly detection use.

SmartRecruit: AI-Powered Recruitment Automation Tool [🔗](#)

Tech stack: Python, Flask, SQL, Scrapy, scikit-learn, OpenAI API, HTML/CSS

- Built a Flask-based web tool to automate candidate screening and evaluation workflows.
- Integrated scraping, SQL storage, and K-Means clustering to organize candidate data and support shortlisting.
- Used the OpenAI API to generate coding test feedback and improve evaluation consistency.

Speechchain – Joint ASR-TTS Research Toolkit

- Implemented a machine speech chain using Speechchain by integrating **ASR** and **TTS** in a closed loop for bidirectional speech and text learning.
- Trained a joint ASR TTS model on the Mini **LibriSpeech** dataset using PyTorch with log Mel feature extraction and TensorBoard monitoring.
- Configured multi-GPU training and dynamic data pipelines to enable scalable and real-time speech transformation.

Cold Email Generation System with Generative AI [🔗](#)

- Engineered an automated email generation platform using Flask, Llama 3.1, and LangChain, demonstrating programming proficiency in Python and advanced NLP techniques relevant to AI applications.
- Executed generative AI workflows with a focus on user-specific context, showcasing a strong foundation in generative models, which aligns with the research scope of the role.

Azure AI-Powered Document Extraction Pipeline [🔗](#)

Tech stack: Python, Azure Durable Functions, GPT-4 Vision, Azure Container Apps, Azure Monitor, Docker, CI/CD

- Built a scalable document extraction pipeline for PDF processing using Azure Durable Functions with batch-based orchestration.
- Used GPT-4 Vision with structured prompting to extract key fields from invoices and convert them into consistent outputs.
- Deployed the service with Docker and Azure Container Apps, with monitoring, logging, retries, and CI/CD to improve reliability and extraction efficiency by 25%.

CNN-Based Emotion Classification from Images [🔗](#)

- Built a **CNN model** in **TensorFlow/Keras** to classify facial expressions (happy vs. sad).
- Preprocessed a custom dataset, trained with the **Adam optimiser**, and evaluated performance with accuracy/loss metrics.
- Deployed the model for inference on unseen images and saved it for future use.

ESPnet – End-to-End Speech Processing Toolkit

Tech stack: Python, ESPnet, PyTorch, Kaldi, LibriSpeech, LJSpeech, NumPy, Pandas, Matplotlib

- Built ASR and TTS pipelines in ESPnet using pretrained Transformer models on LibriSpeech and LJSpeech, fine-tuning them with Kaldi-style recipes.
- Worked with speech translation and voice activity detection to strengthen end-to-end speech processing workflows.
- Automated evaluation using WER and CER plus spectrogram checks, tuning decoding strategies and acoustic outputs.

Fashion Image Generation using Generative Adversarial Networks (GANs) [🔗](#)

- Developed a **GAN-based** model with **TensorFlow** and **Keras** to generate realistic fashion images from the Fashion MNIST dataset, optimizing generator-discriminator training using noise injection and adaptive learning rates for improved image quality and stability.

Recommendation System for E-commerce using Generative AI

- Built a hybrid recommendation engine using **content-based filtering (TF-IDF + cosine similarity)** and **collaborative filtering (SVD)**.
- Processed and analyzed over **50,000 product reviews** to deliver personalized suggestions and improve user engagement.
- Applied **text preprocessing**, feature engineering, and similarity scoring for relevance ranking.
- Demonstrated strong skills in **data cleaning**, model tuning, and user-item interaction analysis.

Real-Time Object Detection and Tracking with YOLOv8 on Custom Dataset

- Built a real-time **object detection** system using **YOLOv8** with high-speed, accurate predictions.
- Trained on a custom-labeled dataset for domain-specific detection.
- Integrated Deep SORT for multi-object tracking across video frames.
- Used **OpenCV** for live video processing and real-time visualization.
- Evaluated using mAP, **precision**, **recall**, and FPS metrics; suitable for surveillance, traffic monitoring, and automation.

Transcriptomics Data Analysis – DESeq2

- Built a reproducible pipeline in R for differential gene expression using DESeq2 on RNA-seq data.
- Normalised and preprocessed high-throughput data; visualised results via PCA, heatmaps, and MA plots.
- Conducted gene ontology analysis for biological interpretation.
- Demonstrated familiarity with molecular data and its clinical applications.

Medical Q&A Chatbot using MedQuAD

Tech stack: Python, LangChain, FAISS, Google Generative AI, Streamlit, Hugging Face embeddings

- Developed a retrieval-augmented chatbot using LangChain and FAISS over the MedQuAD dataset to answer medical questions reliably.
- Integrated Google Generative AI for response generation and Streamlit for the frontend interface.
- Focused on domain-specific context retrieval, answer traceability, and user-friendly deployment

Extract Text from Images using Tesseract OCR

- Developed a **Python-based OCR system** using **Tesseract** for automated text extraction from **images** and **PDFs**.
- Enhanced recognition accuracy through **OpenCV preprocessing** and noise reduction.
- Automated **batch text processing** and structured data output for downstream NLP analysis.

Multilingual Machine Translation API using MarianMT and FastAPI

- Built a multilingual translation API using MarianMT models from Hugging Face, supporting English, German, and French text translation.
- Integrated FastAPI to serve translation requests via REST endpoints with JSON input/output.
- Implemented lightweight tokenization, model caching, and batch translation for high performance in on-premise or cloud environments.

Multi-Agent Communication System (Agent2Agent, A2A Protocol)

- Implemented A2A-based multi-agent communication with structured message schemas, routing, and context propagation for reliable inter-agent coordination.
- Built role-based agent orchestration with task delegation, handoff control, and response aggregation to execute multi-step workflows.

Human-in-the-Loop Autonomous Navigation in CARLA

Tech stack: Python, CARLA, PyTorch, OpenCV, NumPy, Pandas

- Built a CARLA-based autonomous driving prototype with human corrective feedback.
- Used steering and braking interventions to improve model behavior through imitation learning.
- Measured performance using collision rate, route completion, and intervention frequency.

Synthetic Data and Hybrid Simulation Pipeline for AI Model Testing

Tech stack: Python, Unity or Unreal Engine, PyTorch, OpenCV, Label export pipeline, Docker

- Built a simulation-based pipeline to generate labeled data for vision model training and testing.
- Trained and validated models under varying weather, lighting, and scene conditions.
- Evaluated robustness across synthetic and mixed test scenarios.

Bilingual German-English LLM for Student and Workplace Productivity

Python, PyTorch, Hugging Face, SentencePiece, Pandas, Streamlit, Git, GitHub, NLP

- Built an end-to-end LLM pipeline from scratch for bilingual German-English text generation.
- Created data collection, cleaning, deduplication, tokenizer training, and model pretraining workflows.
- Developed a Streamlit demo to convert messy notes into summaries, reports, and meeting minutes.